

Walmart EDA

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```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
-- Attaching packages ----- tidymodels 1.4.0 --

v broom      1.0.9      v rsample     1.3.1
v dials      1.4.2      v tune        2.0.0
v infer      1.0.9      v workflows   1.3.0
v modeldata  1.5.1      v workflowsets 1.1.1
v parsnip    1.3.3      v yardstick   1.3.2
v recipes    1.3.1

-- Conflicts ----- tidymodels_conflicts() --
x scales::discard() masks purrr::discard()
x dplyr::filter()   masks stats::filter()
x recipes::fixed()  masks stringr::fixed()
x dplyr::lag()       masks stats::lag()
x yardstick::spec() masks readr::spec()
x recipes::step()    masks stats::step()

# How many stores does the features data include? What about the training data?
n_distinct(train$Store)
```

```
[1] 45
```

```
n_distinct(features$Store)
```

```
[1] 45
```

```
# looks like they both contain 45 stores, however train has many more rows than features, pr
```

```
length(train$Store)
```

```
[1] 421570
```

```
length(features$Store)
```

```
[1] 8190
```

```
# investigate missing data
```

```
sum(is.na(features$Markdown1))
```

```
[1] 4158
```

```
train %>%  
  count(Store, Dept)
```

```
# A tibble: 3,331 x 3
```

	Store	Dept	n
	<dbl>	<dbl>	<int>
1	1	1	143
2	1	2	143
3	1	3	143
4	1	4	143
5	1	5	143
6	1	6	143
7	1	7	143
8	1	8	143
9	1	9	143
10	1	10	143

```
# i 3,321 more rows
```

o, three issues that I will need to consider when creating a predictive model for this data are:

1. Upon viewing the features and training data, I saw that they have data for the same 45 stores, but not the same dates. The training data has many more rows, because it includes more dates than the features data. I will have to figure out the best way to join them.
2. There is a lot of missing data, for example 4158 rows in the features data have NA values. I will need to decide how to deal with this.
3. Imbalanced data. There are some store/department combos, such as store 1 department 77 that only have a few observations.